Neural Networks 206 (2027) 109801

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

Structure-constrained language-assisted diffusion model for unpaired low-dose computed tomography angiography reconstruction

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ARTICLE INFO

Keywords:
Computed tomography angiography
Unpaired data
Iodinated contrast media
Diffusion model
Language model

ABSTRACT

The application of iodinated contrast media (ICM) improves the sensitivity and specificity of computed tomography (CT) for a wide range of clinical indications. However, high ICM doses can cause problems such as kidney damage and life-threatening allergic reactions. Deep learning methods can generate CT images of normal-dose ICM from low-dose ICM, reducing the required dose while maintaining diagnostic power. However, existing methods struggle to achieve accurate enhancement with incompletely paired images, mainly because of the limited ability of the model to recognize specific structures. To overcome this limitation, we propose a Structure-constrained Language-assisted Diffusion Model (SLDM), a unified medical generation model that integrates structural synergy and spatial intelligence. First, the structural prior information of the image is effectively extracted to constrain the model inference process, thus ensuring structural consistency in the enhancement process. Subsequently, a language-assisted strategy with spatial intelligence is introduced, which integrates the functions of visual perception and spatial reasoning, thus prompting the model to achieve accurate enhancement. Finally, the subtraction angiography enhancement module is applied, which serves to improve the contrast of the ICM region to suitable interval for observation. Qualitative analysis of visual comparison and quantitative results of several metrics demonstrate the effectiveness of our method in angiographic reconstruction for low-dose contrast medium CT angiography.

1. Introduction

Iodinated contrast media (ICM) is applied in computed tomography angiography (CTA) to enhance tissue contrast for evaluating anatomical structures and pathology (Fähling et al., 2017; Yu et al., 2007). It helps improve the sensitivity and specificity of CT diagnosis for various common clinical indications, such as stroke, trauma, and tumors (Chiu & Chu, 2022). However, ICM poses certain risks to patients, including life-threatening allergic reactions such as kidney injury (Golledge & Eagle, 2008; Rubin, 1997). To mitigate the patient health risks associated with ICM in CTA, reducing the dose of ICM is a direct approach. However, the reduction compromises organ contrast by affecting diagnostic performance (Kim et al., 2021; Xie et al., 2021). Recently, deep learning (DL) algorithms were validated to reduce the contrast dose and synthesize virtual contrast enhanced images (McCollough et al., 2015; Zhang et al., 2022a).

Generative adversarial networks (GANs) have been applied to CTA reconstruction (Kang et al., 2019; Lyu et al., 2023; Pang et al., 2023). But existing GAN-based CTA enhancement algorithms still face two key challenges including the model's generalization and stability. First, these models rely on single-step and end-to-end generation, which enables faster reconstruction speeds, but limits the reliable enforcement of constraints during the reconstruction process (Zhu et al., 2017). Second, while GAN-based methods model the target sample distribution through

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https://doi.org/10.1016/j.neunet.2026.109801
Received 4 February 2026; Received in revised form 15 September 2026; Accepted 24 October 2026
Available online 29 October 2026
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generator-discriminator interactions and learn mapping relationships of image pairs. The lack of likelihood evaluation prevents them from fully understanding the physiological features of medical images, thus overlooking the overall organ structure (Dhariwal & Nichol, 2021). The effectiveness of all these methods is inevitably limited by the inherent flaws of GANs.

The score-based generation model (SGM) adopts explicit likelihood-based representations and a stepwise sampling process to improve sample fidelity in generative modeling tasks (Wu et al., 2024, 2023). Unfortunately, there are no relevant studies applying SGM to CTA reconstruction. In-depth analysis, existing SGM-based methods face critical challenges when applied to ICM reconstruction, as shown in Fig. 1. First, the SGM-based method exhibits false negative reconstruction results, leading to unexpected structural distortions due to mismatches between low-dose CTA (LDCT) images and normal dose CTA (NDCT) images during acquisition. This issue is particularly prominent in vascular regions, as shown in Fig. 1(a). Second, the SGM-based method exhibits false positive reconstruction results, where bones that should not be enhanced are incorrectly enhanced, leading to discrepancies with clinical findings, as shown in Fig. 1(b). Third, the stability of SGM-based reconstruction method is usually affected by data contrast differences. For example, when ICM contrast differences of the aortas are significant in LDCT, the reconstruction results are inconsistent and prone to interference, as shown in Fig. 1(c). In summary, addressing the heterogeneous variations induced by ICM is essential for achieving high-quality CTA reconstruction.

To address the above challenges, we propose the Structure-constrained Language-assisted Diffusion Model (SLDM), which establishes three-component collaborative framework of structural prior constraint, language-assisted and dynamic contrast enhancement. First, we propose a structure-constrained mean-reverting stochastic differential equation (SDE). Specifically, our method augments the state space by introducing explicit structural channels. As the intensity components of the SDE diffuse toward potentially misaligned targets, the structural components are constrained to strictly recover to the true structure. Second, we establish a multimodal semantic-guidance mechanism that encodes anatomical descriptors (e.g., target structure type and approximate anatomical locations) as text prompts. These semantic indicators guide the reconstruction process through cross-modal attention, enhancing spatial awareness in specific clinical contexts. Third, the model achieves subtractive contrast enhancement through local contrast adaptation technology. Structural and grayscale information from the enhanced regions is precisely extracted to dynamically correct contrast. As a result, SLDM accomplishes the alignment of specific structures in the complex transformation process. Our main contributions are as follows.

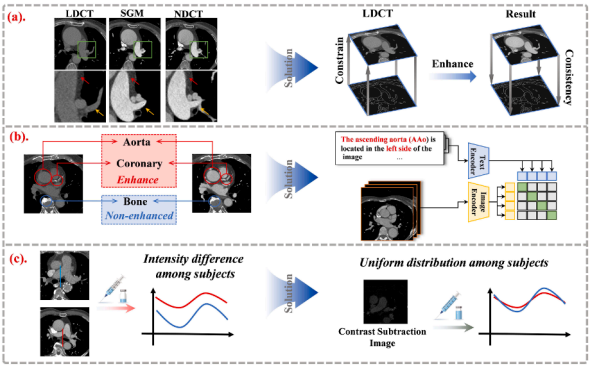


Fig. 1. The proposed method employs three strategies to cope with the corresponding problems. (a) Structural differences in weakly paired data affect reconstructed structures; (b) False enhancements caused by entanglement between enhanced and non-enhanced regions; (c) Intensity differences of subjects affect the reconstructed grayscale.

- We introduce a structure-constrained strategy into the image-to-image diffusion model. By using structure channels as anchors, the network is forced to learn conditional identity mappings of geometry, thereby effectively decoupling texture enhancement from structure preservation on unpaired data.
- A text-prior assisted strategy integrating anatomical information is proposed. By translating semantic information of anatomical structures-such as organ types and spatial attributes-into anatomical priors, this approach effectively enhances the performance of the SLDM.
- The dynamic subtraction angiography enhancement module is designed based on intensity differences. By extracting features of the distribution of contrast agents, locally adaptive contrast guided by dynamic subtraction angiography is used to improve the flexibility of enhancement.
- We demonstrated efficacy through extensive evaluation on clinical datasets, showing superior reconstruction performance compared to competing methods in preserving structural fidelity and improving contrast.

2. Related works

2.1. Virtual contrast enhanced CT algorithm

Existing research primarily focuses on GAN-based methods to improve the image quality of virtually enhanced contrast CT through adversarial learning (Kang et al., 2019; Lyu et al., 2023). For example, Xie et al. (2021) proposed a CycleGAN model to generate contrast-enhanced CT based on simulated non-contrast CT. Pang et al. (2023) proposed two-stage GAN-based synthesizer to perform inter-synthesis between LDCT and NDCT images. Zhu et al. (2017) proposed a Cycle-GAN framework for recovering contrast-enhanced CT from non-contrast simulated CT by redesigning the cycle consistency loss. However, GAN-based methods consider LDCT enhancement as a simple image translation task, so they cannot specifically understand the structural consistency and structural dissimilarity in LDCT enhancement.

2.2. Visual-language models-assisted image generation

Currently, significant advances in visual language models (VLM) like contrastive language-image pre-training (CLIP) are revolutionizing medical imaging (Dalmaz et al., 2022; Mokady et al., 2021; Rombach et al., 2022). It is conceived that they can effectively integrate comprehensive prior knowledge to provide effective information constraints (Chambon et al., 2022; Radford et al., 2021). In the medical field, VLMs have been employed for analyzing chest X-ray slices and related radiology reports (Niu & Wang, 2023). For example, Zhang et al. proposed ConVIRT which learns chest X-ray image representations through bidirectional contrastive learning of image-text pairs, significantly reducing the need for labels in downstream classification tasks (Zhang et al., 2022b). Chen et al. proposed a CT denoising method based on language-engaged dual-space alignment(LEDA) loss, which enhances reconstruction quality by aligning continuous perceptual space with discrete semantic space (Chen et al., 2024). VLMs can be applied as generic interfaces for cross-domain image translation, enabling precise domain transformations with task-specific textual cues to synergize multi-source information and specific tasks (Jin et al., 2024). However, no previous studies have explored the integration of VLMs with the semantic distinction requirements between enhanced regions and non-enhanced regions in CTA reconstruction.

2.3. Preliminaries in image restoration SDE

Image Restoration SDE (IR-SDE) directly simulates the image degradation process via a mean-reverting SDE, which effectively circumvents

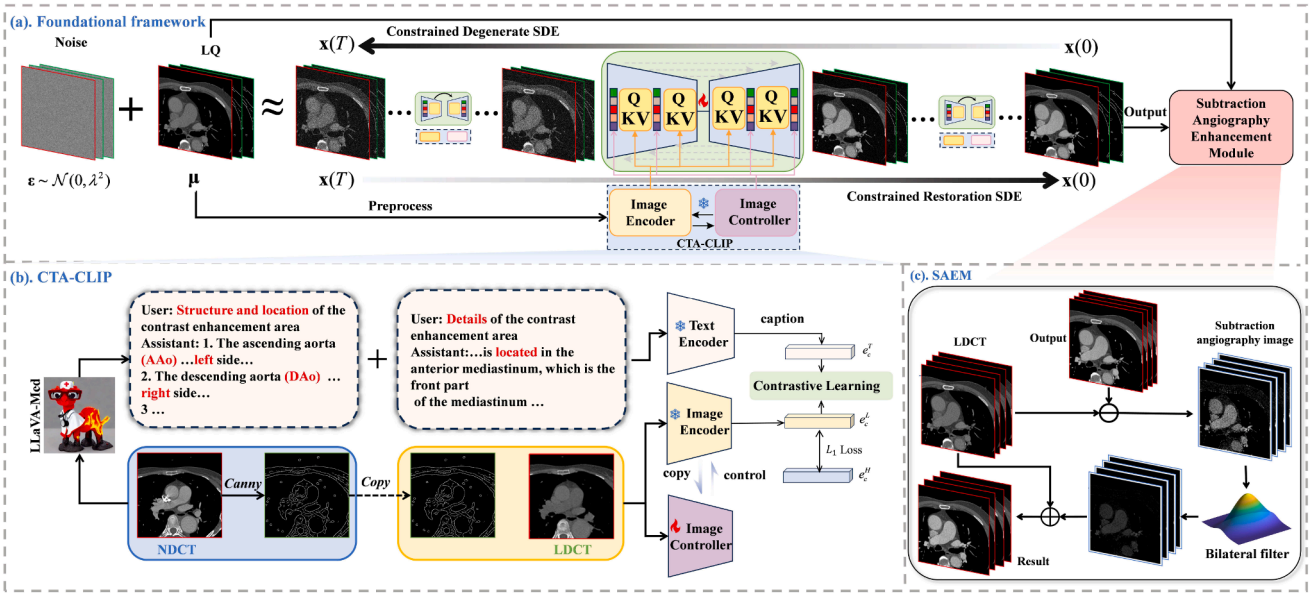


Fig. 2. Overview of the proposed SLDM. (a) IR-SDE serves as the foundational framework, incorporating structural prior constraints and language-assisted guidance during both training and inference to preserve structural fidelity. (b) The pre-trained CTA-CLIP encoder facilitates fine-grained feature extraction and cross-modal semantic alignment. (c) The subtraction angiography enhancement module identifies contrast-enhanced regions and performs adaptive contrast refinement.

the problem that the degradation process is difficult to accurately estimate (Luo et al., 2023a, 2024b). During the diffusion process, for any state $t \in [0, T]$, IR-SDE is defined as an Ornstein-Uhlenbeck process (Yue et al., 2024). The forward and reverse diffusion processes can be expressed by the following equations:

$$\begin{cases} d\mathbf{x} = \theta_t(\mu - \mathbf{x})dt + \sigma_t d\omega \\ d\mathbf{x} = [\theta_t(\mu - \mathbf{x}) - \sigma_t^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})]dt + \sigma_t d\hat{\omega} \end{cases} \tag{1}$$

where μ is the state mean, and θ_t and σ_t are positive parameters that vary with time t, and they describe the mean-reversion speed and stochastic volatility in the diffusion process, respectively. ω is a standard Wiener process (Arjovsky & Bottou, 2017), and $\hat{\omega}$ is a reverse-time Wiener process. $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is the score function, which can be approximated by training a time dependent neural network s_ϕ. In addition, the maximum likelihood learning process can also be used to optimize the inverse process. The optimal inversion state $\mathbf{x}_{t-1}^*$ at step $t-1$ can be naturally obtained from $\mathbf{x}_t$ via the most negative log-likelihood:

$$\mathbf{x}_{t-1}^* = \underset{\mathbf{x}_{t-1}}{\operatorname{argmin}}[-\log q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)], \tag{2}$$

where $\mathbf{x}_{t-1}^*$ denotes the ideal state obtained in the reverse direction from $\mathbf{x}_t$. By solving the above objective, the score prediction network is trained with maximum likelihood loss, which specifies the optimal backward path $\mathbf{x}_{t-1}^*$ for all times (Liu et al., 2024):

$$\mathbf{x}_{t-1}^* = \frac{1-e^{-2\bar{\theta}_{t-1}}}{1-e^{-2\bar{\theta}_t}} e^{\theta_t'}(\mathbf{x}_t - \mu) + \frac{1-e^{-2\theta_t'}}{1-e^{-2\bar{\theta}_t}} e^{-\bar{\theta}_{t-1}}(\mathbf{x}_0 - \mu) + \mu, \tag{3}$$

where $\theta_t' = \int_{t-1}^t \theta_t dt$ and $\bar{\theta}_t = \int_0^t \theta_t dt$. Based on Eq. (3), $\mathbf{x}_{t-1}^*$ is obtained using $\mathbf{x}_t$ during diffusion. Hence, IR-SDE can be optimized through the following training target:

$$L_\gamma(\phi) = \sum_{t=1}^T \gamma_t \mathrm{E}\left[\left\|\mathbf{x}_t - \mathbf{x}_{t-1}^* - (d\mathbf{x}_t)_{s_\phi(\mathbf{x}_t, \boldsymbol{\mu}, t)}\right\|\right], \tag{4}$$

where γ_t represents the weight at time t, and $s_\phi(\mathbf{x}_t, \boldsymbol{\mu}, t)$ denotes the score of state $\mathbf{x}_t$ under condition μ at time t. After training, IR-SDE can utilize $s_\phi(\mathbf{x}_t, \boldsymbol{\mu}, t)$ to iteratively generate high-quality image via numerical solver, starting from a noisy initial state $\mathbf{x}_T$.

3. Methods

3.1. Overview of the proposed SLDM

The overall workflow of SLDM is depicted in Fig. 2 and consists of three main phases: the diffusion model prior generation phase, the CTA-CLIP assistance phase, and the subtraction angiography enhancement phase. In the prior generation phase (Fig. 2(a)), structure prior constraint is introduced to extract spatial information to maintain precise structural consistency during the enhancement process. In the CTA-CLIP assistance phase (Fig. 2(b)), language-assisted mechanism enables the perception of morphological features across different anatomical structures. Subsequently, the cross-attention mechanism constrains the backward diffusion process, precisely enhancing contrast regions through text prompts specific to particular structures. In the subtraction angiography enhancement phase (Fig. 2(c)), the structural information of the contrast agent regions is accurately extracted in the subtraction angiography enhancement module (SAEM). These regions are dynamically optimized to achieve more flexible and adaptive reconstruction result.

The overall SLDM for the training and testing process is outlined in Algorithm 1. The training process consists of two stages: fine-tuning of CTA-CLIP and training of the structure prior constrained IR-SDE. The core of CTA-CLIP is to use contrastive learning to align the semantic representations of NDCT images with the image features of LDCT. Structural prior constrained IR-SDE is trained using LDCT-NDCT data pairs, while the structure prior and text information of NDCT are used as constraints. During the testing process, LDCT images are gradually enhanced by leveraging the structural prior constraints and language assistance score functions learned during training and the optimal reverse-time path. Finally, SAEM further refines the reconstructed NDCT images.

3.2. Structure constrained IR-SDE

Although NDCT and LDCT obtained in clinical settings may come from the same patient and the same equipment, patient displacement or respiration can cause mismatches between NDCT and LDCT. This results in significant structural distortions in the image-to-image diffusion model (Liu et al., 2023; Luo et al., 2023a; Shi et al., 2024). We consider two image domains: NDCT domain $\mathbf{x} \in \mathcal{X}$ and LDCT domain $\mathbf{y} \in \mathcal{Y}$. The

Algorithm 1: SLDM: training and sampling process.

Data: LDCT data, NDCT data
Input: LDCT image: $\mathbf{y}$, NDCT image: $\mathbf{x}$, structural prior: $\mathbf{s}$, hyper-parameters: $\boldsymbol{\sigma}_t$, $(\theta_t)_{t\in[1,T]}$, T, λ
Output: Enhanced image: $\mathbf{x}_{out}$

I. Training Process

A: CTA-CLIP Model Pre-training

Get text description by LLaVA-Med: (NDCT data) $\rightarrow c_{Text}$;
Generate text and image embedding: $\{e_c^T, e_c^L\}$;
Use Eq. 9 to calculate the contrastive learning loss ;

B: Structure-constrained IR-SDE Training

Obtain score function: s_ϕ // The score obtained by combining structural prior constraints and language assistance ;

II. Testing Process

Data initialization: $\tilde{\mathbf{x}}(t) = [\mathbf{x}(t), \mathbf{s}(t)]$;
for $t = T$ ***down to*** 0 **do**
  $\theta_i' := \int_{i-1}^{i} \theta_t dt,\ \bar{\theta}_t = \int_0^t \theta_j dj,\ \sigma_t^2/\theta_t = 2\kappa^2,\ v_t = \kappa^2(1 - e^{-2\bar{\theta}_t})$ // Calculate required parameters ;
  $\hat{\mathbf{x}}_0 = e^{\bar{\theta}_t}(\tilde{\mathbf{x}}_t - \mu_t - \sqrt{v_t}\mathbf{s}_\phi) + \mu_t$ // Estimate initial value ;
  $\tilde{\mathbf{x}}_{t-1}^* = \frac{1-e^{-2\bar{\theta}_{t-1}}}{1-e^{-2\bar{\theta}_t}} e^{\theta_t'}\left(\tilde{\mathbf{x}}_t - \mu\right) + \mu + \frac{1-e^{-2\theta_t'}}{1-e^{-2\bar{\theta}_t}} e^{-\bar{\theta}_{t-1}}\left(\hat{\mathbf{x}}_0 - \mu\right)$;
  $\tilde{\beta}_t = \frac{(1-e^{-2\bar{\theta}_{t-1}})(1-e^{-2\theta' t})}{1-e^{-2\bar{\theta}_t}}$;
  $p(\tilde{\mathbf{x}}_{t-1}|\tilde{\mathbf{x}}_t, \hat{\mathbf{x}}_0) = \mathcal{N}(\tilde{\mathbf{x}}_{t-1}|\tilde{\mathbf{x}}_{t-1}^*, \tilde{\beta}_t\mathbf{I})$ // Posterior distribution;
end

C: Subtraction angiography enhancement

$\mathbf{x}_{fill} = \hat{\mathbf{x}}_0, \mathbf{x}_{mask} = \mathbf{y}$;
Compute subtraction: $\mathbf{x}_{sub} = \mathbf{x}_{fill} - \mathbf{x}_{mask}$;
Apply filter: $\mathbf{x}_{sub}^B = \text{Bilateral}(\mathbf{x}_{sub})$;
Final output: $\mathbf{x}_{out} = \mathbf{x}_{mask} + \lambda\mathbf{x}_{sub}^B$;
return $\mathbf{x}_{out}$

imagess in these domains are unpaired. So the goal is to learn a restoration function that maps the LDCT to NDCT while preserving the structure of the LDCT. We introduce a structure extraction function C, where $C(\mathbf{x})$ represents the structural information of any image $\mathbf{x}$. During the training phase, we construct extended training pairs:

$$\tilde{\mathbf{x}}_0 = \left[\mathbf{x}_0, C(\mathbf{x}_0)\right], \tilde{\boldsymbol{\mu}} = \left[\mathbf{y}_0, C(\mathbf{x}_0)\right]. \tag{5}$$

We consider an extended state $\tilde{\mathbf{x}}(t) = [\mathbf{x}(t), \mathbf{s}(t)]$, where $\mathbf{x}(t)$ is image part, and $\mathbf{s}(t)$ is structure part. Therefore, we propose the forward and backward processes of structure constraint IR-SDE as follows:

$$\begin{cases} d\tilde{\mathbf{x}} = \theta_t(\tilde{\boldsymbol{\mu}} - \tilde{\mathbf{x}})dt + \sigma_t d\omega \\ d\tilde{\mathbf{x}} = \left[\boldsymbol{\theta}_t(\tilde{\boldsymbol{\mu}} - \tilde{\mathbf{x}}) - \boldsymbol{\sigma}_t^2 \nabla_{\tilde{\mathbf{x}}} \log p_t(\tilde{\mathbf{x}})\right]dt + \boldsymbol{\sigma}_t d\tilde{\boldsymbol{\omega}} \end{cases} \tag{6}$$

For the image channel, we optimize the network to match the optimal reverse solution Eq. (3) under the maximum likelihood learning Eq. (2). For the structural channel, the optimal reverse solution is:

$$\mathbf{s}_{i-1}^* = \frac{1-e^{-2\bar{\theta}_{i-1}}}{1-e^{-2\bar{\theta}_i}} e^{-\theta' i}(\mathbf{s}_i - \mathbf{s}_{\text{gt}}) + \frac{1-e^{-2\theta' i}}{1-e^{-2\bar{\theta}_i}} e^{-\bar{\theta}_{i-1}}(\mathbf{s}_{\text{gt}} - \mathbf{s}_{\text{gt}}) + \mathbf{s}_{\text{gt}}, \tag{7}$$

where $\mathbf{s}_{\text{gt}} = C(\mathbf{x}_0)$. If $\mathbf{s_i} = \mathbf{s}_{\text{gt}}$, then $\mathbf{s}_{i-1}^* = \mathbf{s}_{\text{gt}}$. Therefore, during training, when the initial mean of $\mathbf{s}(T)$ is $\mathbf{s}_{\text{gt}}$, the network learns to maintain the structural channels as $\mathbf{s}_{\text{gt}}$.

During the testing phase, the input to the structural channels is changed to $\mathbf{s}_{\text{lq}} = C(\mathbf{y}_0)$.

Theorem (Consistency of generated results): Let the image generated during the testing phase be $\hat{\mathbf{x}}_0$, and the corresponding structural component be $\mathbf{s}_0$. Then, when the backward step number $T \rightarrow \infty$:

1. $\lim_{T\to\infty} \mathbb{E}\left[\left\|\mathbf{s}_0 - \mathbf{s}_{\text{lq}}\right\|_2\right] = 0.$
2. There exists a constant $K > 0$ such that:

$$\mathbb{E}\left[\left\|C(\tilde{\mathbf{x}}_0) - \mathbf{s}_{\text{lq}}\right\|_2\right] \leq K\mathbb{E}\left[\left\|\mathbf{s}_0 - \mathbf{s}_{\text{lq}}\right\|_2\right] + \epsilon_{\text{align}}. \tag{8}$$

The proof is provided in Section A.1. The above theorem shows that the structure of the generated result is constrained by the structure of the test input. Ultimately, by integrating structural prior constraints, we adapt IR-SDE to scenarios with unpaired data. This approach enhances LDCT images while ensuring structural fidelity.

3.3. CTA-CLIP assistance phase

We pre-train CTA-CLIP with text descriptions of structural information from NDCT and image information from LDCT. The core components of CTA-CLIP include a text encoder based on Transformer and an image encoder based on Vision Transformer. These components process low-dose CT images into embeddings that can be matched with corresponding textual (Radford et al., 2021). For text embedding, we employ LLAVA-med which is a medical large language model (LLM) developed by Microsoft to generate corresponding text description features (Li et al., 2023). Specifically, LLAVA-med analyzes the structure of NDCT images and obtains specific feature descriptions, including organ types and locations. Then CTA-CLIP converts the image information from LDCT into image feature vectors by feeding it into an image encoder. Similarly, the text description information obtained from NDCT is encoded into text feature vectors by feeding it into a text encoder.

To align text and image embeddings, we calculate the cosine similarity between the embedding pairs though cross-entropy loss (Luo et al., 2024b). The loss can be simplified as follows:

$$\mathcal{L}_c(a, b) = -\log\left(\frac{e^{\cos(a,b^+)}}{\sum e^{\cos(a,b^*)}}\right), \tag{9}$$

where a represents LDCT image, b^+ represents the text of the matched positive sample, b^* represents all texts including texts of unmatched negative samples, cos denoting cosine similarity. By maximizing the probability of the similarity of positive samples among the similarities of all sample pairs, CTA-CLIP can achieve cross-modal image-semantic alignment. To optimize the embedding of textual and image information, we train the model using the following objective function:

$$\mathcal{L}_c(\zeta) = \mathcal{L}_c(e_c^L, e_c^T; \zeta), \tag{10}$$

where e_c^L represents the image-based embeddings derived from LDCT, and e_c^T represents the structural description derived from NDCT. ζ represents the learnable parameters of the controller. CTA-CLIP enables precise enhancement of localized structures during LDCT enhancement by learning to align image information from LDCT with corresponding text embeddings from NDCT.

After completing the pre-training of CTA-CLIP, a cross-attention module is introduced to integrate the text descriptions and prompt functions into the reverse enhancement process (Niu & Wang, 2023). The output of the text encoder of feature descriptions can guide the diffusion model to generate images that conform to the text descriptions. The underlying mechanism lies in transforming the diffusion model into a more flexible conditional image generator. To preprocess textual cues from linguistic cues c_{Text}, a structure-specific encoder is introduced that projects c_{Text} to an intermediate representation, which is then passed through the implementation of

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V, \tag{11}$$

where Q (Query), K (Key), V (Value) are learnable projection matrices, d is the task token dimension. In the cross-attention mechanism, Q is generated from image features in the U-Net, while K and V are generated from semantic vectors of the text encoder. Through the cross-attention mechanism, the image features in the diffusion model enhancement process will be aligned with the corresponding text information, thereby being guided by the text information during the generation process.

3.4. Subtraction angiography enhancement module

Inspired by digital subtraction angiography (DSA) technology, we designed the SAEM algorithm based on image differences to adapt to individual vascular characteristics. SAEM designates LDCT as the mask image $\mathbf{x}_{mask}$, while considering the enhanced image $\mathbf{x}_0$ as the filling image $\mathbf{x}_{fill}$. As depicted in Fig. 2(c), SAEM acquires the subtracted image by computing the disparity between the mask image and the filling image. In subtraction images, bilateral filtering eliminates redundant noise interference, retaining only the vascular images containing contrast agent.

First, the subtraction angiography image $\mathbf{x}_{sub}$ is obtained from the mask image $\mathbf{x}_{mask}$ and the filling image $\mathbf{x}_{fill}$. Then, bilateral filtering is applied to the subtraction angiography images to remove noise, yielding high-quality subtraction angiography images $\mathbf{x}^B_{sub}$. Finally, the high-quality subtraction angiography image is fused with LDCT by weighting parameter λ, which obtains accurate reconstructed image $\mathbf{x}_{out}$ and enables adjustable contrast enhancement.

$$\begin{cases} \mathbf{x}_{sub} = \mathbf{x}_{fill} - \mathbf{x}_{mask}, \\ \mathbf{x}^B_{sub} = \text{Bilateral}(\mathbf{x}_{sub}), \\ \mathbf{x}_{out} = \mathbf{x}_{mask} + \lambda \mathbf{x}^B_{sub}. \end{cases} \tag{12}$$

Thus, SAEM confers on SLDM the superiority of the ability to achieve controllable grayscale adjustment targeting the enhancement region. The benefit of this simple module is obvious and meaningful. It allows physicians the flexibility to customize vascular enhancements rather than the traditional controlled-dose regimen based on the patient's pathology.

4. Experiments

4.1. Datasets

4.1.1. CTA dataset acquisition and pre-processing

Datasets were acquired from aortography-enhanced CTA scans at Beijing Anzhen Hospital of Capital Medical University. For each subject, both NDCT and LDCT scans were performed, covering the anatomical range from the lung apices to the lesser trochanter of the femur. All clinical tests were approved by the Medical Ethics Committee of Beijing Anzhen Hospital (Ethics approval number: 2022 Kelun Review No. (26), approval date: December 28, 2022). NDCT examinations were carried out on a 256-slice multi-detector system (Revolution CT, GE Healthcare, United States). Iodixanol 320 (320 mg I/ml) was administered intravenously at a volume of 40-70 ml. Tube potential was tailored to patient BMI, ranging from 80 to 120 kV, while tube current was automatically adjusted with CareDose 4D technology. The acquisition yielded an in-plane spatial resolution of 0.3-0.5 mm and a nominal slice thickness of 0.625 mm. LDCT scans used the same volume of a 1:2 mixture of iodixanol 320 and saline, with scanning parameters matching those of the NDCT protocol except for the iodinated contrast dose. The dataset comprises normal-dose and low-dose CTA scans from 55 patients, 40 patients were used for training, 5 patients for validation and 10 patients for testing. Each patient provided 224 CT slices for the study. All subsets include images covering critical regions (aorta, coronary arteries, pulmonary veins), avoiding bias toward specific body parts. The dataset was split at the patient level to prevent information leakage between the training, validation, and test sets. It should be noted that due to the limitations of multiple acquisitions, the NDCT is not ground truth image of the LDCT. Therefore, since the LDCT-NDCT data pairs are not perfectly matched, NDCT can only serve as a reference image for LDCT.

To enhance visualization of arterial-tissue intensity differences, windowing adjustments were applied to both LDCT and NDCT datasets. NDCT showed a theoretical maximum arterial enhancement of 700 HU. A window level of 200 HU and width of 1000 HU were empirically set, with intensities outside this range truncated: values below -300 HU were reset to -300 HU, and those above 700 HU to 700 HU. Both LDCT and NDCT datasets underwent identical truncation and were normalized to a scale of 0–1 for network input.

4.1.2. Preparation of structure prior dataset

A carefully designed structured dataset of prior information had been constructed to maintain precise structural constraints. A three-channel data composition approach was employed which consists of one image channel and two structural-prior channels. The canny operator was used to generate Structure prior. For the training datasets, the first channel of each input pair used grayscale images from both NDCT and LDCT, while the remaining channels utilized structure prior of NDCT. For the testing datasets, the first channel relied solely on LDCT grayscale images, with the remaining channels constructed from structure of LDCT. The uniquely designed dataset enables structural constraints for the enhancement process.

4.1.3. Text dataset construction

The language-assisted datasets were constructed using NDCT images, and corresponding text descriptions (Fig. 3(a)). Text descriptions were obtained through LLAVA-med-v1.5-mistral-7b which was a medical LLM developed by Microsoft (Li et al., 2023). Owing to the capacity limitation of LLM, they cannot accurately respond to questions when multiple cue words were input simultaneously. To improve the model's ability to understand question details, we employed a multi-semantic prompt strategy. Specifically, we designed two prompts leveraging the LLM's contextual information processing capabilities. The first prompt requested a description of structures in the contrast-enhanced CT section. The second prompt asked for information on the location of the contrast-enhanced area. A text file was generated from these prompt words, offering a detailed summary of structural information on anatomical structures such as the aorta, coronary artery, and the corresponding locations in the NDCT images. Additionally, it provided information about the location of the contrast agent regions (as shown in Fig. 3(b).) Next, image-text embeddings were produced by leveraging the pre-trained encoders within CTA-CLIP. Specifically, CTA-CLIP's image and text encoders adopt ViT-B-32, a Vision Transformer variant, as their shared backbone.

4.2. Evaluation metrics

The evaluation involves the state-of-the-art image translation methods, including MUNIT (Huang et al., 2018), IdentityGAN (Kang et al., 2019), Cycle-GAN (Zhu et al., 2017), and MALAR (Zhang et al., 2022a). These studies have been widely applied in the field of CTA reconstruction. Due to the current lack of diffusion model-based CTA reconstruction algorithms, we applied two diffusion model algorithms designed for

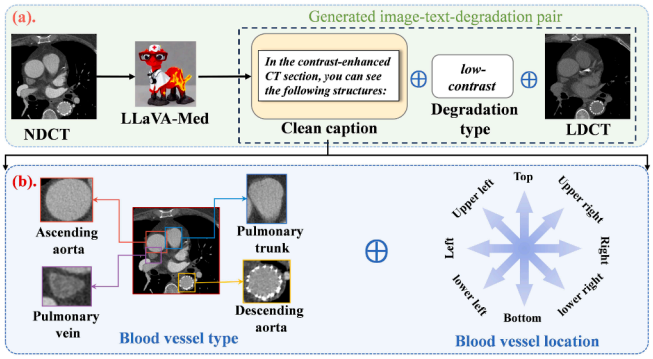


Fig. 3. Construction of the semantic dataset for SLDM. (a) An example of generating an anatomical text description from an NDCT image using LLaVA-Med and forming the image-text information used for CTA-CLIP training; (b) an example of semantic information describing vessel types and approximate anatomical locations.

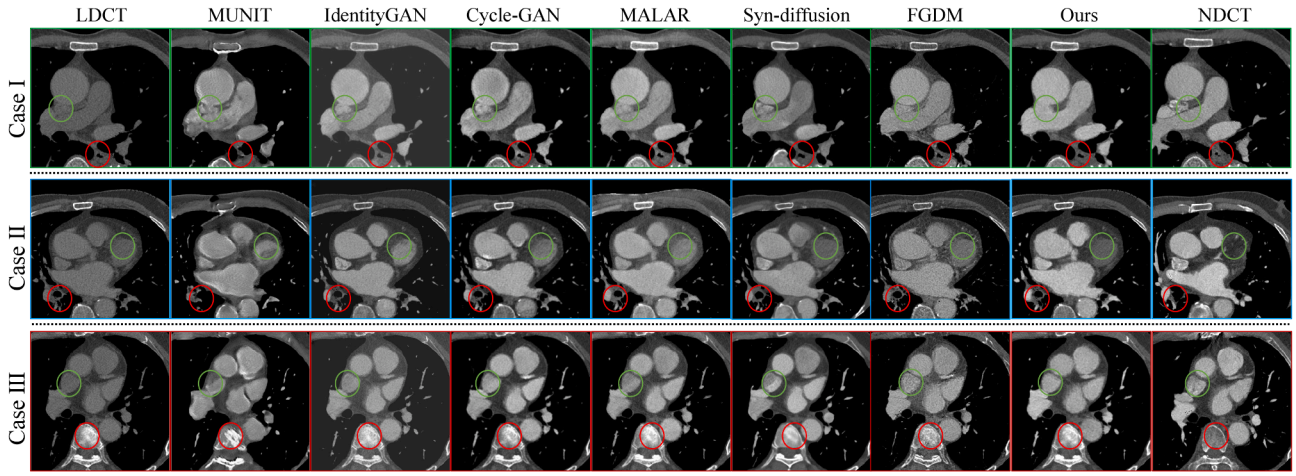


Fig. 4. The visualization comparison between our method and state-of-the-art methods. Synthesized images from competing methods are displayed along with LDCT and NDCT (reference) images for representative tasks. The display window for the reconstructed image is set to [-300, 700] HU. The portion of the image in the green and red circles is the region of interest (ROI). Compared to baselines, proposed method maintains higher anatomical fidelity.

Table 1
Patient-level quantitative comparisons of different methods.

Method	PSNR ↑	SSIM ↑	SNR ↑	LPIPS ↓	ISNR ↑
MUNIT	16.3648 ± 0.3673*	0.5469 ± 0.0584*	6.0867 ± 0.6944*	0.2522 ± 0.0229*	−3.9392 ± 1.4916*
IdentityGAN	20.5670 ± 1.5238*	**0.7421 ± 0.0813**	10.2888 ± 1.1393*	0.1598 ± 0.0442*	0.2630 ± 0.0988*
Cycle-GAN	19.8162 ± 2.2075*	0.6616 ± 0.1097*	9.5380 ± 1.9040*	**0.1005 ± 0.0560**	−0.4878 ± 1.2716*
MALAR	21.6173 ± 1.4186*	0.7087 ± 0.0757*	11.3392 ± 1.1478*	0.1443 ± 0.0462*	1.3133 ± 0.8089*
Syn-diffusion	20.9733 ± 1.4592*	0.6997 ± 0.0692*	10.6951 ± 1.3077*	0.1605 ± 0.0391*	0.6693 ± 0.9819*
FGDM	17.8007 ± 1.8387*	0.5261 ± 0.0871*	7.6112 ± 1.7565*	0.2226 ± 0.0559*	0.4606 ± 0.4721*
SLDM	**22.4289 ± 1.9754**	0.7376 ± 0.0837	**12.1508 ± 1.6698**	0.1305 ± 0.0468	**2.1249 ± 1.0815**

Results are reported as patient-level mean ± standard deviation over 10 test patients. Higher PSNR, SSIM, SNR, and ISNR values indicate better performance, whereas lower LPIPS values indicate better performance. The best results are shown in bold, and the second-best results are underlined. * indicates a statistically significant difference compared with SLDM according to a two-sided paired t-test ($p < 0.05$).

image translation tasks as the control group for this study. Syn-diffusion (Özbey et al., 2023) and FGDM (Li et al., 2024), medical image translation framework derived from diffusion model, were modified and applied to LDCT reconstruction.

To quantitatively evaluate the reconstruction performance, the NDCT images were first registered to the corresponding LDCT images using ANTs (Murphy et al., 2011) and were treated as clinical references rather than strict ground truth. Peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), signal-to-noise ratio (SNR), improvement in signal-to-noise ratio (ISNR), and learned perceptual image patch similarity (LPIPS) were subsequently calculated based on the registered image pairs. Higher PSNR, SSIM, SNR, and ISNR values indicate better reconstruction performance, wher-eas a lower LPIPS value indicates greater perceptual similarity to the reference image. ISNR quantifies the reconstruction improvement relative to the original LDCT image, with a positive value indicating effective restoration.

To investigate the generalizability of various reconstruction methods to clinically realistic LDCT images, we randomly selected reconstructed images for subjective evaluation. All subjects were anonymous and randomly shuffled. One radiologist with 20 years of experience independently evaluated the randomized and anonymized images in two sessions separated by a washout interval of 48 weeks. Evaluation criteria included image sharpness, noise level, contrast, and detail preservation. We used an integer score range from 1 to 5, with higher scores indicating better quality. Specifically, a score of 5 indicates good contrast, good vessel circling, few artifacts, and easy diagnosis; a score of 1 indicates unexplained contrast, severe artifacts, indistinguishable vessel circling, and no diagnosis.

4.3. Training details

Spatial structure was used as conditions in the training process. The forward and backward diffusion processes of the model were supervised by the spatial structure. We used the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.99$. The total training steps were fixed to 700 000 and the initial learning rate set to 10^{-4} and was halved every 200 000 iterations. The noise level σ and the number of diffusion denoising steps T were fixed to 50 and 100. The stationary variance λ^2 was set to 10. The batch size was set to 16. Our model was implemented using PyTorch.

4.4. Experimental results

4.4.1. Comparison with the competing methods

Fig. 4 shows the experiments comparing the reconstruction results for three different CT slices. MUNIT is an unsupervised model and can be observed to exhibit significant structural errors as well as contrast differences in the structures. IdentityGAN exhibits global shadowing and anomalous enhancements in local regions, as shown by the red and green ROIs. This is due to its inability to recognize specific structures. Due to the cycle-consistency constraint, Cycle-GAN maintained relatively good spatial structure; however, its ability to model content information remained limited, resulting in nonuniform intensity distributions. MALAR shows decent reconstruction of local structure and contrast due to its combination of multiscale local features and long-range dependency, but still has undesired contrast deficits in some regions. Syn-diffusion is a diffusion-based medical image translation method; however, the highlighted ROIs show insufficient

preservation of both structural and contrast information. In comparison, FGDM has more consistent structure and gray scale values. Unfortunately, its internal structure has more pronounced noise, which can interfere with physician diagnosis. SLDM addresses these limitations through synergistic integration of structure constraints and multi-semantic guidance, achieving superior boundary accuracy and contrast fidelity. The subtraction angiography enhancement module further enables clinically appropriate intensity calibration within optimal HU ranges, enhancing both quantitative performance and diagnostic utility.

The quantitative results in Table 1 show that the proposed method achieved the highest PSNR, SNR, and ISNR, together with the second-best SSIM and LPIPS. Although IdentityGAN yielded a slightly higher SSIM and Cycle-GAN achieved a slightly lower LPIPS, neither difference was statistically significant compared with the proposed method ($p = 0.055$ and $p = 0.100$, respectively); therefore, these numerical differences do not provide sufficient evidence that the competing methods performed better. In contrast, the proposed method demonstrated statistically significant improvements in PSNR, SNR, and ISNR over all competing methods, as well as significant advantages in SSIM and LPIPS over the remaining applicable comparisons. For more detailed statistical metrics, see Table A.3 in the appendix. Because the NDCT images registered using ANTs served as clinical reference images rather than strict ground truth, small differences in reference-based metrics should still be interpreted cautiously. Accordingly, the quantitative findings were considered together with visual comparisons, blinded radiologist assessments, automated segmentation results, and subsequent anatomical measurements, which collectively support the effectiveness of the proposed method. The computational complexity of the diffusion-based methods is further analyzed in Section A.2.

4.4.2. Quantitative blind evaluation results

Fig. 5 presents the aggregated scores across two blinded evaluation sessions and the mean scores. SLDM achieved the highest mean score in both sessions, demonstrating consistently favorable subjective image quality. The two assessments also demonstrated excellent intra-rater agreement, with an exact agreement rate of 86.7%, a quadratic-weighted Cohen's κ of 0.953, and an absolute-agreement intraclass correlation coefficient of 0.953 (Koo & Li, 2016; Landis & Koch, 1977). Although MALAR and Syn-diffusion achieved favorable results in some cases, SLDM showed the most consistently high subjective scores. Nevertheless, because the evaluation was performed by a single radiologist, further validation involving multiple independent radiologists is required to establish inter-rater reliability.

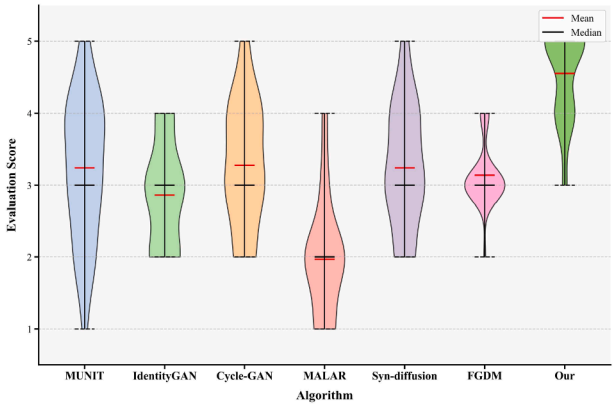


Fig. 5. Violin plots of subjective image-quality scores obtained from two independent blinded evaluation sessions conducted by the same radiologist. The red and black lines denote the mean and median, respectively.

4.4.3. Segmentation experiment

To evaluate vascular structure preservation without relying on the unpaired NDCT reference, TotalSegmentator (Wasserthal et al., 2023) was applied to the original LDCT images and the reconstructed images. Fig. 6 illustrates the segmentation results of different algorithms for the aorta, heart, superior vena cava, and pulmonary veins. The regions indicated by yellow arrows in the 2D slices and red arrows in the 3D views highlight the shortcomings of the comparative algorithms regarding segmentation details. The segmentation obtained from each LDCT input was used as the structural reference because the reconstructed image retained the same anatomical coordinate system. Dice coefficient, intersection over union (IoU), centerline Dice (clDice), and average symmetric centerline distance (ASCD) were used to evaluate vascular overlap, topology preservation, and centerline consistency. As shown in Table 2, the proposed method achieved competitive Dice, IoU, and clDice values and the lowest ASCD, indicating high consistency with the LDCT-derived reference segmentation during contrast enhancement.

4.4.4. Anatomical measurements

To further assess clinically relevant aortic morphology, AortaExplorer (Paulsen et al., 2026) was used to measure aortic attenuation, ascending aortic volume, regional cross-sectional areas, vessel diameter ranges, and tortuosity indices , as shown in Table 3. Cross-sectional area profiles along the vessel centerline were also compared between the registered NDCT clinical reference and the reconstructed result. As shown in Fig. 7, the two profiles showed similar variation along the aorta, while the differences in regional area, diameter, volume, and tortuosity remained limited, indicating that the proposed method increased vascular attenuation without substantially altering aortic morphology.

4.5. Ablation study

To investigate the validity of our proposed method, ablation experiments were carried out on various components and hyperparameters. We performed three different ablation experiments to validate the effectiveness of the structure constraint module: without structure constraint, the structure of single-channel and the structure of LDCT. In addition, the validity of the CTA-CLIP and SAEM were verified. All ablation experiments were conducted on the same dataset.

4.5.1. Effectiveness of structure constraint

As shown in Fig. 8(a), to systematically evaluate the effectiveness of structure spatial module, we conducted comparative experiments between the proposed framework and the IR-SDE baseline architecture. The results demonstrate that while both approaches achieve contrast enhancement in specific regions confirming effective image domain transformation. But the unconstrained IR-SDE method exhibits significant structural aberrations. Through analysis, we verified that integrating structure constraints substantially improves structural preservation by maintaining boundary region connectivity.

Further validation of our dataset construction strategy was performed through systematic comparison of different image-channel configurations (Fig. 8(b)). It is important to note that the ablation experiments in Fig. 8(a) exhibit results closer to NDCT. In fact, it defeats the original purpose of the ICM enhancement algorithm, i.e., the vessel contrast is enhanced but the structure remains unchanged. In contrast, our method exhibits more faithful and reasonable results.

Experimental trials contrasting single-channel grayscale with two-channel contour combination against dual-channel grayscale with single contour configurations revealed superior local structure recovery in the latter paradigm (Fig. 8(b)). The objective evaluation indicators presented in Fig. 8(b) show similar agreement. This empirical evidence supports our hypothesis that reduced cross-domain uncertainty in single image pair configurations enhances reconstruction fidelity through more deterministic feature mapping.

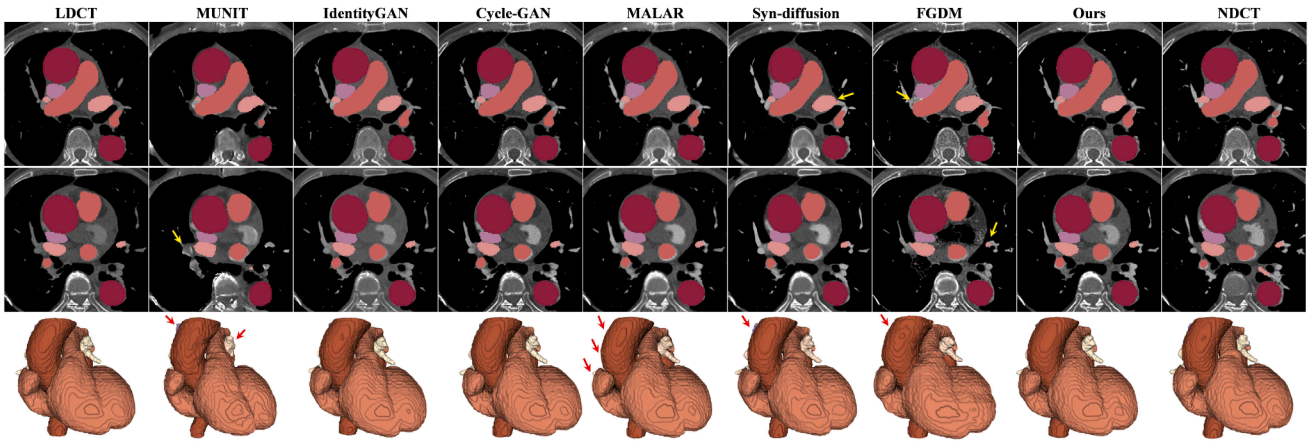


Fig. 6. Segmentation results between our method and different comparative methods.

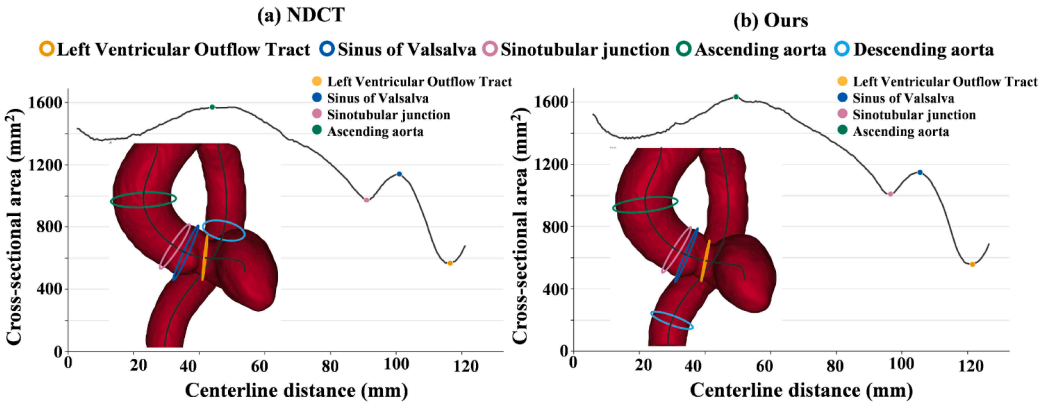


Fig. 7. Comparison of vessel cross-sectional area distribution along the vessel centerline.

Table 2
Quantitative segmentation metrics for different methods.

Metric	MUNIT	IdentityGAN	CycleGAN	MALAR	Syn-diffusion	FGDM	Ours
Dice (%) ↑	66.58	**98.51**	97.95	96.56	93.17	95.40	98.28
IoU (%) ↑	49.90	**97.07**	95.99	93.35	87.21	91.21	96.62
clDice (%) ↑	78.65	**98.24**	96.77	95.62	96.41	97.19	97.26
ASCD (mm) ↓	3.28	1.16	1.19	1.38	1.38	1.12	**1.10**

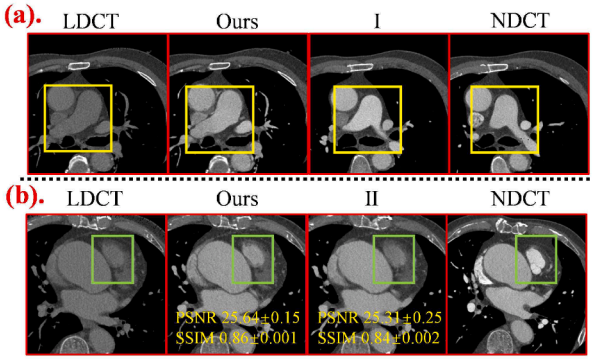


Fig. 8. Comparison results from different types of structure datasets. (a) Comparison results from the IR-SDE base framework (I) and our method; (b) Comparison results obtained with the structure of single-channel (II) and our method.

4.5.2. *Effectiveness of CTA-CLIP*

The effectiveness of the CTA-CLIP was evaluated through comparative analyses of reconstruction performance with and without textual constraints, as demonstrated in Fig. 9. Comparative analysis reveals that the model without CTA-CLIP exhibits unintended enhancement during the contrast optimization process (as shown in Fig. 9 (a) red and white arrow), which may compromise diagnostic reliability. Integration of text-based language-assisted effectively regularizes the enhancement trajectory, significantly improving both spatial accuracy and intensity precision in critical anatomical regions. The three-dimensional volume rendering analysis (Fig. 9 (b)) further confirms this improvement.

4.5.3. *Robustness analysis of structural priors*

To further evaluate the robustness of the proposed structure-constrained diffusion process against imperfect structural priors, we introduced controlled perturbations to the LDCT-derived edge maps by randomly removing 5% to 25% of the edge pixels. As illustrated in Fig. 10, although the reconstruction fidelity slightly degrades in tandem with the corrupted priors (indicated by yellow arrows), the core anatomical structures remain highly consistent with minimal visual discrepancies. Quantitative metrics indicate that randomly removing 25% of the structures results in a decrease of approximately 0.41 dB in PSNR and 0.0124 in SSIM. These findings demonstrate that the diffusion model does not merely replicate the input priors; rather, it effectively

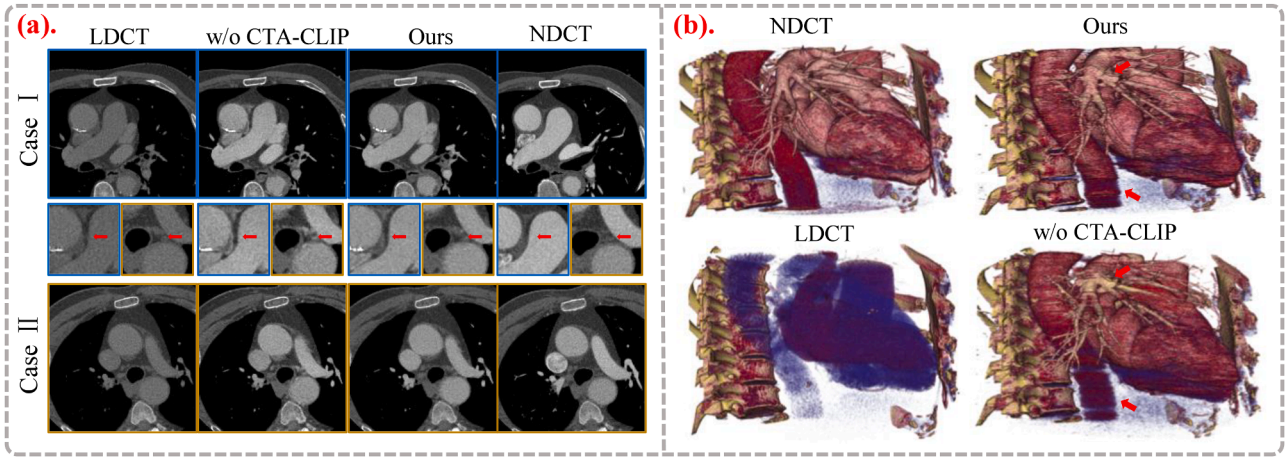


Fig. 9. A comparative analysis of reconstruction results with and without CTA-CLIP. (a) illustrates the reconstruction outcomes for various slices, (b) provides a three-dimensional visualization of the overall anatomical structure.

Table 3
Comparison of anatomical measurements.

Metric	NDCT	Ours	Difference
Mean Aortic HU	273.61	327.62	+54.01
Median Aortic HU	280	343	+63
AAo Volume (cm^3)	154.43	161.36	+4.49%
LVOT Area (mm^2)	567.25	557.00	–1.81%
SoV Area (mm^2)	1142.50	1149.75	+0.63%
STJ Area (mm^2)	973.50	1010.25	+3.78%
Max AAo Area (mm^2)	1570.75	1634.25	+4.04%
LVOT Diameter Range (mm)	23.69–31.75	23.54–30.37	1.38 (Max)
SoV Diameter Range (mm)	34.94–42.52	36.05–42.50	1.11 (Max)
STJ Diameter Range (mm)	33.84–37.79	33.54–39.27	1.48 (Max)
AAo Diameter Range (mm)	43.49–46.51	44.10–47.67	1.16 (Max)
AAo Tortuosity Index	1.312	1.367	+4.16%
DAo Tortuosity Index	1.148	1.167	+1.71%

AAo: Ascending Aorta, DAo: Descending Aorta,
LVOT: Left Ventricular Outflow Tract,
SoV: Sinus of Valsalva, STJ: Sinotubular Junction.

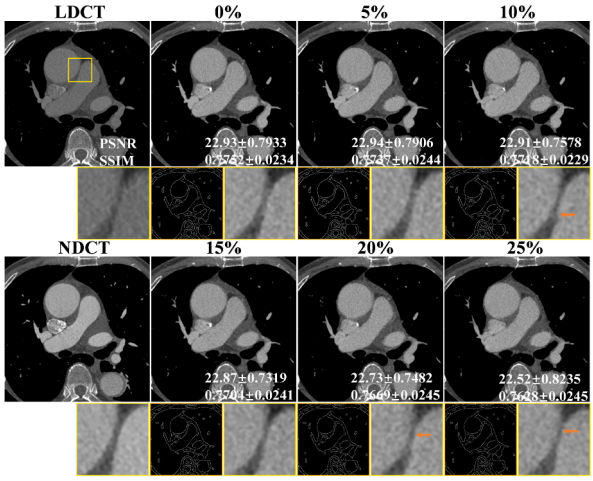


Fig. 10. Comparison of reconstruction results under varying degrees of structural loss. The bottom-right panel shows an enlarged view of the pale yellow ROI, while the bottom-left panel displays contours with varying degrees of edge truncation (highlighted by orange boxes). Quantitative metrics for the different levels of structural loss are presented in the bottom-right panel.

synergizes the incomplete structural guidance with its learned anatomical distribution to reconstruct robust anatomical details.

4.5.4. *Effectiveness of SAEM*

The SAEM's performance was evaluated through parametric comparison of SLDM with different weighting configurations, as illustrated in Fig. 11. Experimental results demonstrate effective extraction of subtraction angiography features from reconstructed images, enabling precise modulation of contrast ranges in enhanced regions. As evidenced in Fig. 12, this functionality allows controlled adjustment of intensity intervals through parameter weighting, significantly improving reconstruction flexibility while maintaining anatomical fidelity. The parametric adaptability of our SAEM demonstrates the advantages in balancing contrast optimization with structural preservation in various clinical scenarios.

5. Discussion

5.1. *Uniqueness of the SLDM*

While recent bridge-based diffusion models (Chung et al., 2023; Liu et al., 2023; Shi et al., 2024) demonstrate potential in learning the mapping between distributions, they predominantly necessitate either fully paired datasets or explicit knowledge of the degradation operator. Notably, our proposed method offers a streamlined and effective integration strategy that has the potential to extend these frameworks to the regime of unpaired data.

Parallel to these developments, conditional generative architectures such as ControlNet (Zhang et al., 2023) and OminiControl (Tan et al., 2025) leverage structural priors (e.g., edges) to guide image synthesis. However, adapting these methods to the current task presents two fundamental challenges. First, the lack of pixel-wise alignment between LDCT and NDCT data during fine-tuning precludes the use of the LDCT intensity distribution as a data-fidelity constraint. Consequently, the inference process relies exclusively on the extracted LDCT contours. Given that low-dose contours are inherently susceptible to noise artifacts, this unconstrained reliance risks error augmentation. Second, these paradigms typically depend on large-scale foundational models (e.g., Stable Diffusion) adapted via complex fine-tuning procedures, introducing significant computational overhead. In contrast, our approach provides a theoretically grounded, and data-efficient solution tailored for real-world medical deployment.

DA-CLIP leverages large-scale pre-trained vision language models as a universal framework for image restoration (Luo et al., 2024b). Inspired by this approach, All-Inclusive Image Enhancement (AIIE) integrates knowledge of various degradation types into its model, enabling it to handle multiple image degradation tasks (Ju et al., 2025). Dual-

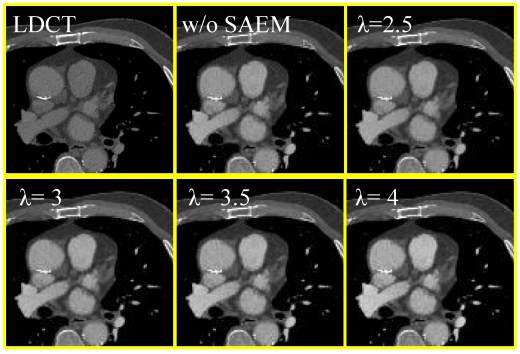


Fig. 11. The comparison of results for different parameters λ in the SAEM.

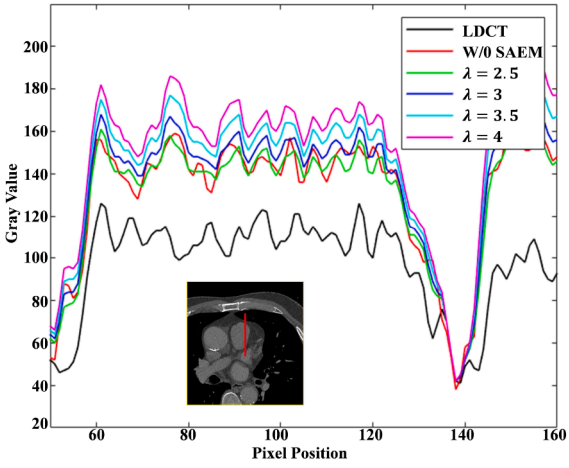


Fig. 12. The gray scale histogram of SAEM-free and different weighting parameters λ.

Domain CLIP-Assisted Residual Optimization Perception Model (DuDoCROP) fine-tunes DuDoCLIP via contrastive learning in both image and sinogram domains (Zhang et al., 2024). This process extracts semantic descriptions to guide diffusion model optimization. Although these studies provide valuable guidance, CTA reconstruction is a unique task which focuses specifically on enhancing specific structures (like coronary artery). SLDM innovatively acquires medical categories of key features and encodes them as text embeddings. This effectively guides the enhancement of specific structures, offering a novel approach for CTA reconstruction.

5.2. Clinical benefits of SLDM

Excessive use of iodinated contrast media (ICM) can cause allergic reactions and acute kidney injury. This problem is particularly prominent in high-risk patients, such as those with kidney disease and the elderly. The SLDM can directly generate enhanced images with equivalent normal-dose quality from low-dose ICM images. This provides a safer alternative for CTA examinations. Training our model does not require fully paired data, which greatly reduces the difficulty of clinical deployment. Through precise structural constraints and text prompts, the SLDM specifically addresses structural distortions and abnormal enhancements in enhancement processes. In addition, SAEM provides doctors with a personalized optimization scheme. In the future, integrating SLDM into clinical imaging systems is expected to become a safe and efficient tool.

5.3. Limitations of the SLDM

Although SLDM has advanced low-dose CTA reconstruction, there are four key limitations that still warrant investigation. This section delineates each limitation and proposes prospective research directions. (i) Lack of real-time capability: On a CT image of 384×384 pixel, SLDM requires about 7.2s, outperforming conventional diffusion models yet remaining slower than GAN-based approaches. Achieving real-time reconstruction will likely necessitate integrating consistency model or other single-step generative paradigms. (ii) Although ANTs registration reduces spatial mismatch, the registered NDCT images remain clinical references rather than strict ground truth. Therefore, marginal differences in PSNR and SSIM should be interpreted cautiously. (iii) All data in this study were acquired at a single institution using the same scanner model, which may limit the generalizability of the proposed method across institutions, scanner vendors, and acquisition protocols. Future studies will perform external validation on larger multi-center and cross-vendor datasets and investigate domain adaptation strategies to improve scanner robustness.

5.4. Future directions

In addition to addressing the current limitations, several challenging directions merit further investigation. First, future studies could extend generative priors to CTA with extremely low or even no contrast-agent administration (Zhou et al., 2023), where weak vascular signals may substantially increase reconstruction uncertainty. Second, inspired by medical world models (Yang et al., 2025), future frameworks could model the spatiotemporal dynamics of contrast-agent transport by incorporating patient-specific hemodynamic information, moving beyond static image enhancement. Third, self-evolving medical agents (Almansoori et al., 2025) may enable adaptive restoration through experience-based memory, autonomous reflection, and closed-loop prompt optimization according to patient anatomy and acquisition conditions. Such developments may support task-adaptive restoration across heterogeneous clinical cohorts, although expert supervision and prospective multi-center validation will remain essential.

6. Conclusion

In conclusion, we proposed a CTA reconstruction diffusion-based CTA enhancement model for enhancing contrast in low-dose ICM. SLDM is generated using structure prior constraints across domains, which effectively maintains structural consistency. Then, language-assisted mechanism was introduced, which effectively alleviated the problem of abnormal enhancement of local structures. The subtraction angiography enhancement module is designed to adjust contrast. SLDM enhances vascular contrast in low-contrast-dose CTA while maintaining the major anatomical structures of the input images. The complementary quantitative, segmentation-based, anatomical, and subjective evaluations support its potential, although larger multi-center validation and task-specific diagnostic assessment remain necessary.

CRediT authorship contribution statement

Genyuan Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization; **Zihao Wang:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis; **Zhifan Gao:** Writing – review & editing, Supervision, Resources, Project administration; **Lei Xu:** Software, Resources, Data curation, Conceptualization; **Zhen Zhou:** Resources, Investigation, Conceptualization; **Haijun Yu:** Visualization, Validation, Software; **Jianjia Zhang:** Validation, Supervision, Software, Data curation; **Xiujian Liu:** Software, Investigation, Data curation; **Weiwei Zhang:** Investigation, Data curation; **Shaoyu**

Wang: Writing – review & editing, Data curation; **Huazhu Fu:** Writing – review & editing; **Fenglin Liu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation; **Weiwen Wu:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Key Research and Development Program of China under Grant (2022YFA1204201, 2022YFF0706400); in part by the National Natural Science Foundation of China under Grant (82271986, 82402256, 62171067, 62471502, 62471501, 92570107); in part by Shenzhen Science and Technology Program under Grant ZDCY20251230090545009 and Grant JCYJ20240903101311015; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2026B1515020019. The authors would like to thank all the funding agencies for their financial support.

Appendix A.

A.1. The proof of consistency of generated results

During the training phase, the network learns the conditional score function $\nabla_{\mathbf{s}} \log p_t(\mathbf{s}(t) \mid \mathbf{s}_{\text{gt}}) = -\frac{\mathbf{s}(t)-\mathbf{s}_{\text{gt}}}{v_t}$, where $v_t = \lambda^2(1-e^{-2\bar{\theta}_t})$ (Luo et al., 2023a). During the testing phase, the input structure is $\mathbf{s}_{\text{lq}}$. Assuming the network possesses generalization capabilities, for any conditional structure $\mathbf{s}_{\text{cond}}$, the predicted score is approximated as $-\frac{\mathbf{s}(t)-\mathbf{s}_{\text{cond}}}{v_t}$. Therefore, the drift term for the structural channel in the reverse SDE is given by:

$$\frac{d\mathbf{s}}{dt} = \theta_t(\mathbf{s}_{\text{lq}} - \mathbf{s}) - \sigma_t^2\left(-\frac{\mathbf{s}-\mathbf{s}_{\text{lq}}}{v_t}\right) = -\left(\theta_t + \frac{\sigma_t^2}{v_t}\right)(\mathbf{s}-\mathbf{s}_{\text{lq}}). \tag{A.1}$$

This constitutes an Ornstein-Uhlenbeck process (Maller et al., 2009) with an equilibrium point at $\mathbf{s}_{\text{lq}}$. As $T \to \infty$ and $\int_0^T(\theta_t + \sigma_t^2/v_t)dt \to \infty$, we have $\mathbb{E}[\mathbf{s}(0)] \to \mathbf{s}_{\text{lq}}$ and the variance approaches zero. Thus:

$$\lim_{T\to\infty} \mathbb{E}[||\mathbf{s}_0 - \mathbf{s}_{\text{lq}}||_2] = 0. \tag{A.2}$$

Let G denote the image generation function of the network, with input $[y_{\text{test}}, \mathbf{s}_{\text{lq}}]$ and output $\hat{x}_0$. Define $G_s([y,s]) = C(G([y,s]))$ as the structure of the generated image. During training, for a training sample (y, s_{gt}), we have $G_s([y, s_{\text{gt}}]) = s_{\text{gt}}$ (perfect alignment). Assume that G_s is Lipschitz continuous, meaning there exists $L_G > 0$ such that:

$$|G_s([y,s_1]) - G_s([y,s_2])|_2 \le L_G|s_1 - s_2|_2. \tag{A.3}$$

During testing, the input structure is s_{lq}, and the structure of the generated image is $C(\hat{x}_0) = G_s([y_{\text{test}}, s_{\text{lq}}])$. Consider the following decomposition:

$$\begin{aligned}||C(\hat{\mathbf{x}}_0) - \mathbf{s}_0||_2 &= ||G_s([\mathbf{y}_{test}, \mathbf{s}_{lq}]) - \mathbf{s}_0||_2 \\ &\le ||G_s([\mathbf{y}_{test}, \mathbf{s}_{lq}]) - G_s([\mathbf{y}_{test}, \mathbf{s}_0])||_2 \\ &\quad + ||G_s([\mathbf{y}_{test}, \mathbf{s}_0]) - \mathbf{s}_0||_2\end{aligned} \tag{A.4}$$

The first term is bounded by Lipschitz continuity: $\le L_G\|\mathbf{s}_{\text{lq}} - \mathbf{s}_0\|_2$. The second term represents the alignment error: training requires that when the input structure is s_0, the generated image structure should be s_0. Thus, $\|G_s([y_{\text{test}}, \mathbf{s}_0]) - \mathbf{s}_0\|_2 \le \epsilon_{\text{align}}$, where ϵ_{align} is the alignment error during training. Therefore:

$$||C(\hat{x}_0) - s_0||_2 \le L_G||\mathbf{s}_0 - \mathbf{s}_{\text{lq}}||_2 + \epsilon_{\text{align}}. \tag{A.5}$$

Table A.1
Comparison of inference time and model parameter counts.

Method	Inference Time (s)	Model Parameters (Million)	
FGDM	410.2	61.4	
Syn-diffusion	0.4	DiffusionAtoB	28.6
Ours	7.2	DiffusionBtoA	28.6
		Diffusion_Module	49.0
		CLIP_module	246.2
Total Parameters	-	FGDM	61.4
		Syn-diffusion	57.2
		Ours	295.2

Now, combining this with the triangle inequality:

$$\begin{aligned}&||C(\hat{x}_0) - \mathbf{s}_{\text{lq}}||_2 \le ||C(\hat{x}_0) - \mathbf{s}_0||_2 + ||\mathbf{s}_0 - \mathbf{s}_{\text{lq}}||_2 \le \\ &(1 + L_G)||\mathbf{s}_0 - \mathbf{s}_{\text{lq}}||_2^2 + \epsilon_{\text{align}}.\end{aligned} \tag{A.6}$$

Taking the expectation and letting $K = 1 + L_G$, we obtain:

$$\mathbb{E}[||C(\hat{x}_0) - \mathbf{s}_{\text{lq}}||_2] \le K\mathbb{E}[||\mathbf{s}_0 - \mathbf{s}_{\text{lq}}||_2^2] + \epsilon_{\text{align}}. \tag{A.7}$$

As $T \to \infty$, $\mathbb{E}[\|\mathbf{s}_0 - \mathbf{s}_{\text{lq}}\|_2] \to 0$, hence $\mathbb{E}[\|C(\hat{x}_0) - \mathbf{s}_{\text{lq}}\|_2] \le \epsilon_{\text{align}}$. If training is perfect ($\epsilon_{\text{align}} = 0$), then $C(\hat{x}_0)$ is exactly equal to $\mathbf{s}_{\text{lq}}$. In the general case, the error is controlled by $K\mathbb{E}[\|\mathbf{s}_0 - \mathbf{s}_{\text{lq}}\|_2]$.

A.2. Computational complexity of the method

To comprehensively evaluate the practical deployment efficiency of each diffusion-based method, we conducted a statistical analysis of the inference time and model parameter count of diffusion-based approaches. The results are shown in Table A.1. The inference time was measured on a single NVIDIA RTX 4090D GPU with a batch size of 1 and an input size of 384×384. For diffusion-based methods, the measurement includes the complete reverse sampling process, with 4, 2000, and 100 sampling steps for Syn-diffusion, FGDM, and our method, respectively. In terms of inference time, our proposed method requires only 7.2 seconds, achieving a 57× speedup compared to FGDM (410.2 seconds) while maintaining superior generation quality. This significant improvement is primarily attributed to our optimized diffusion sampling strategy. Although Syn-diffusion achieves the fastest inference speed (0.4 seconds), its generation quality is relatively limited and falls short of practical application requirements. Regarding model parameter count, our method totals 295.2M parameters (49.0M + 246.2M), with the CLIP module accounting for the majority of parameters and providing robust semantic understanding capabilities. In comparison, FGDM's bidirectional diffusion architecture has a total of 61.4M parameters, while Syn-diffusion contains 57.2M parameters. Overall, our method achieves a better balance between inference efficiency and model performance, ensuring high-quality generation results while remaining feasible for practical deployment.

A.3. The rationale of the text prior

To verify the plausibility of the generated text content, we added an independent quantitative assessment of the generated prompts. 50 representative axial CTA slices were randomly selected from ten different patients, resulting in 400 vessel-presence evaluations because eight predefined vessel candidates were assessed per slice. Eight predefined great-vessel candidates were evaluated: "ascending aorta, aortic arch, descending thoracic aorta, main pulmonary artery, right pulmonary artery, left pulmonary artery, superior vena cava, and inferior vena cava." Two multimodal foundation models, ChatGPT 5.6 and Gemini 3.1, neither of which was involved in LLaVA-Med prompt generation or SLDM training, independently assessed the image-text consistency. For each slice, the evaluators first determined vessel visibility from the CT image and then compared the generated text with the image-based

Table A.2
Independent quantitative evaluation results for generated prompts.

Metric	ChatGPT	Gemini
Vessel precision	67.2%	63.5%
Vessel recall	89.6%	93.0%
Vessel F1-score	76.8%	75.5%
Relaxed location precision	76.7%	77.5%
Joint entity-location recall	68.8%	72.1%
Joint entity-location F1-score	58.9%	58.5%

assessment. As shown in Table A.2, vessel presence was evaluated using precision, recall and F1-score. Because the textual locations are intended to provide coarse spatial guidance, we adopted a relaxed location criterion. A location description was considered correct when it identified the correct broad side or anatomical region and did not contradict the actual anterior-posterior, superior-inferior, or hilar relationship. For example, describing the ascending aorta as being in the anterior mediastinal region was accepted as a coarse approximation, whereas describing the descending thoracic aorta as being in the anterior mediastinum was considered incorrect because it contradicts its posterior location. The study further utilized the evaluation results to examine consistency between the models, specifically, the proportion of cases where the two models yielded consistent evaluations. The resulting raw consistency was 90.8%.

These results provide a more nuanced interpretation of the generated texts. The high vessel recall of approximately 90% indicates that most visible great vessels were covered by the generated prompts. This broad coverage is desirable for an auxiliary semantic prior because it reduces the likelihood that a relevant vascular structure is completely omitted. However, the moderate precision indicates that the prompts sometimes overpredict vessels that are not visible on the current axial slice. The relaxed location precision of approximately 77% shows that most correctly identified vessels were assigned to a broadly reasonable anatomical region. This automated assessment is a preliminary consistency analysis and does not replace expert radiologist validation, which remains a limitation of the present study.

A.4. Comparing segmentation and text priors

To further investigate the roles of text priors and segmentation priors within the proposed framework, we utilize a pre-trained segmentation model (TotalSegmentator Wasserthal et al., 2023) to obtain anatomical masks and categorize them into coarse anatomical classes. The anatomical prior encoder used the same embedding dimension as CTA-CLIP and its output was injected into the same cross-attention layers. All restoration-network settings, training epochs, and loss weights were kept unchanged; only the prior encoder and its teacher signal were replaced. These segmentation maps serve solely as teacher signals for training the anatomical prior encoder via representation alignment. During the reconstruction inference phase, the system does not take segmentation masks as input; instead, it provides the reconstruction model only with the predicted anatomical embeddings extracted from the LDCT images.

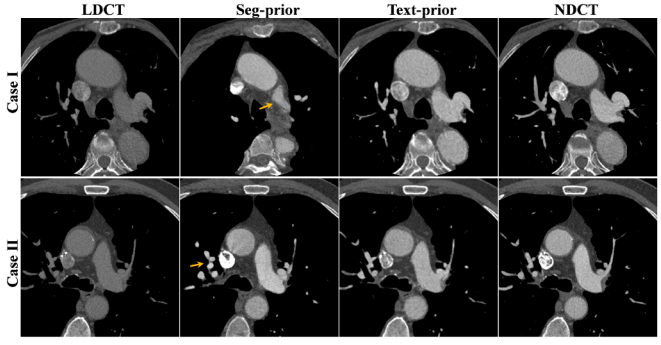


Fig. A.1. Comparison of reconstruction results using segmentation priors and text priors.

Table A.3
Patient-level statistical comparisons between the proposed method and competing methods.

Metric	Comparator	$t(9)$	p value	Hedges' g_z
PSNR	MUNIT	8.942	< 0.001	2.585
	IdentityGAN	5.563	< 0.001	1.608
	Cycle-GAN	5.097	< 0.001	1.474
	MALAR	2.776	0.022	0.803
	Syn-diffusion	5.232	< 0.001	1.513
	FGDM	12.993	< 0.001	3.757
SSIM	MUNIT	14.346	< 0.001	4.148
	IdentityGAN	−2.202	0.055	−0.637
	Cycle-GAN	3.472	0.007	1.004
	MALAR	7.696	< 0.001	2.225
	Syn-diffusion	5.305	< 0.001	1.534
	FGDM	9.078	< 0.001	2.625
SNR	MUNIT	8.942	< 0.001	2.585
	IdentityGAN	5.563	< 0.001	1.608
	Cycle-GAN	5.097	< 0.001	1.474
	MALAR	2.776	0.022	0.803
	Syn-diffusion	5.232	< 0.001	1.513
	FGDM	12.993	< 0.001	3.757
ISNR	MUNIT	8.942	< 0.001	2.585
	IdentityGAN	5.563	< 0.001	1.608
	Cycle-GAN	5.097	< 0.001	1.474
	MALAR	2.776	0.022	0.803
	Syn-diffusion	5.232	< 0.001	1.513
	FGDM	12.993	< 0.001	3.757
LPIPS	MUNIT	−12.566	< 0.001	−3.633
	IdentityGAN	−7.221	< 0.001	−2.088
	Cycle-GAN	1.832	0.100	0.530
	MALAR	−9.619	< 0.001	−2.781
	Syn-diffusion	−6.736	< 0.001	−1.948
	FGDM	−7.483	< 0.001	−2.164

Statistical comparisons were performed using two-sided paired t-tests based on patient-level measurements from 10 test patients. Hedges' g_z was used as the paired effect size with a small-sample correction. For all metrics, the paired difference was defined as Ours minus Comparator. Therefore, positive t and g_z values favor the proposed method for PSNR, SSIM, SNR, and ISNR, whereas negative values favor the proposed method for LPIPS. Because SNR and ISNR differ from PSNR only by patient-specific reference terms that are identical across reconstruction methods, their paired between-method differences, and consequently their t-statistics and effect sizes, are identical to those of PSNR.

As shown in the results in Fig. A.1, under the current experimental setting, the performance of this method falls short of CTA-CLIP, primarily due to significant structural distortions, as indicated by the yellow arrows. Regarding quantitative metrics, the results guided by text-prior guidance achieved a PSNR of 22.93 ± 0.79 and an SSIM of 0.78 ± 0.02, whereas the results guided by segmentation achieved a PSNR of 15.08 ± 0.68 and an SSIM of 0.56 ± 0.06. We further analyzed the different behaviors of segmentation-based and language-guided priors within the current SLDM framework. Our model aims to preserve the geometric characteristics of LDCT through explicit structural constraints while simultaneously learning the contrast transformation between the LDCT and NDCT domains. Consequently, the prior information should provide additional semantic guidance without introducing inaccurate structural correspondences. Directly aligning LDCT images with segmentation representations generated from NDCT data risks introducing structural bias, as the model would be driven to match a discrete anatomical label

space rather than preserving the original LDCT geometry. In contrast, language-guided anatomical descriptions offer soft semantic constraints. They characterize anatomical identities and spatial relationships without mandating pixel-level correspondence, such as vessel types and approximate locations. Thus, the model can acquire anatomical knowledge while maintaining the original structural consistency ensured by the structure-constrained diffusion framework. This aligns with our previous language-based ablation studies, which demonstrated that CTA-CLIP is not merely an alternative method for incorporating anatomical masks; rather, its advantage lies in providing a flexible, soft semantic representation that complements the structure-preservation mechanisms within our unpaired reconstruction framework.

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